

Supporting Online Material

Materials and Methods

The laser beams used to pump the ^{87}Rb vapor are derived from the output of a frequency-stabilized continuous-wave Ti:sapphire laser, which is spatially filtered using a single-mode optical fiber with angled ends, and collimated to a spot size ($1/e$ field radius) of $470\text{ }\mu\text{m}$. The input power of one of the beams (denoted as the forward beam) has a power of $630\text{ }\mu\text{W}$ and the other (denoted as the backward beam) has a power of $250\text{ }\mu\text{W}$. Polarizing beam splitters are placed in each beam so that the input beams are linearly polarized with parallel polarizations.

The isotopically-enriched rubidium vapor ($>90\%$ ^{87}Rb) is contained in a 5-cm-long glass cell heated to $70\text{ }^{\circ}\text{C}$ (corresponding to an atomic number density of $\sim 7 \times 10^{11}\text{ atoms/cm}^3$). The cell is tilted with respect to the incident laser beams to prevent possible oscillation between the uncoated windows. The cell has no paraffin coating on the interior walls to prevent depolarization of the ground-state coherence, nor does it contain a buffer gas that would slow diffusion of atoms out of the pump laser beams. The Doppler-broadened linewidth of the transition at this temperature is $\sim 550\text{ MHz}$. To prevent the occurrence of magnetically-induced instabilities, we use a Helmholtz coil to cancel the ambient magnetic field component along the direction of the counterpropagating laser beams. We believe it is not necessary to use an isotopically-enriched rubidium vapor; it is used as a matter of convenience.

The linearly and co-polarized pump-laser beams counterpropagate through the ^{87}Rb vapor and induce an instability that generates light in the orthogonal state of polarization (Fig. S1). The instability arises from mirror-less self-oscillation, which

occurs when the parametric gain due to nonlinear wave-mixing processes becomes formally infinite. Physically, this implies that new beams of light are spontaneously generated and emitted from the vapor (*S1, S2*).

The instability-generated output light is an emergent spatial structure in the plane orthogonal to the propagation direction. This structure is referred to as a transverse optical pattern (*S3*). In general, the mechanism that gives rise to the pattern formation is the combination of nonlinearity and coupling between the different spatial points due to diffraction (*S3*). Specifically, the pattern forming instability in our system can be interpreted in terms of either optical pumping or four-wave mixing processes (*S4*).

Several experiments have investigated instabilities that occur when laser beams counterpropagate through an atomic vapor and demonstrated spontaneous formation of various transverse optical patterns, including rings and multi-spot off-axis patterns in agreement with our observations (*S4-S6*). A complete theoretical treatment of our experimental system is complicated because the absorption and saturation of the nonlinearity due to optical pumping, the vector nature of the electric field, and the atomic motion are all potentially important. Although no full treatment of this system has been given in the existing literature, several theories predict the general features observed in our experiment, such as the emission of light along a cone and the formation of stable transverse optical patterns with 2- and 6-fold symmetry (*S7-S10*).

In our switching experiment we use a two-spot transverse optical pattern that is stationary in the sense that the transverse spatial structure of the pattern is static, in the absence of a switching beam, over a period of many minutes. Drift of the pattern on this long time scale is primarily due to drift in the laser frequency or pump-beam alignment.

To characterize the instability in greater detail, we measure the total power of the output light as a function of the pump laser frequency. We observe several sub-Doppler features in the power emitted from the vapor cell due to the instability (in the state of polarization orthogonal to that of the input pump beams) as the laser frequency is scanned through the $^{87}\text{Rb } 5S_{1/2} \leftrightarrow 5P_{3/2}$ transition (Fig. S2). The maximum power emitted in the orthogonal polarization occurs when the laser frequency ν is tuned ~ 25 MHz to the high-frequency side of the $5S_{1/2} (F=1) \leftrightarrow 5P_{3/2} (F'=1)$ transition, where F (F') denotes the total angular momentum of the hyperfine level of the ground (excited) state. We tune the laser frequency 5 MHz to the high-frequency side of this peak during the switching experiments.

We believe that it is possible to reduce the switching energy to the single-photon level through some combination of the following: adjusting the vapor length, reducing the size of both pump beams and the switching beam, adding a buffer gas, or using a vapor cell with paraffin-coated walls.

Supporting References and Notes

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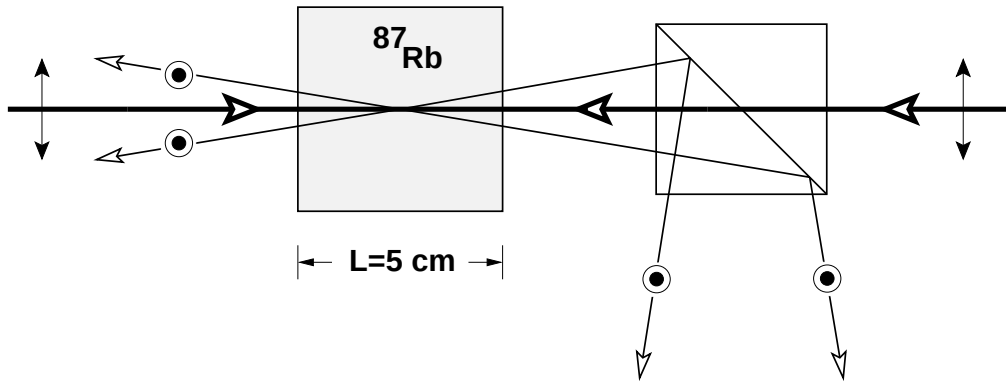


Figure S1: Mirror-less parametric self-oscillation generates new beams of light (*i.e.* a transverse optical pattern). Two pump beams (bold) counterpropagate through a 5 cm ^{87}Rb vapor cell. The pump beams have identical linear polarization (shown as arrows in the plane of the page). The generated light is emitted off-axis in both forward and backward directions with a state of polarization that is linear and orthogonal (shown as circled dots) to that of the pump beams. The polarizing beam splitter (shown on the right) separates generated light from pump light.

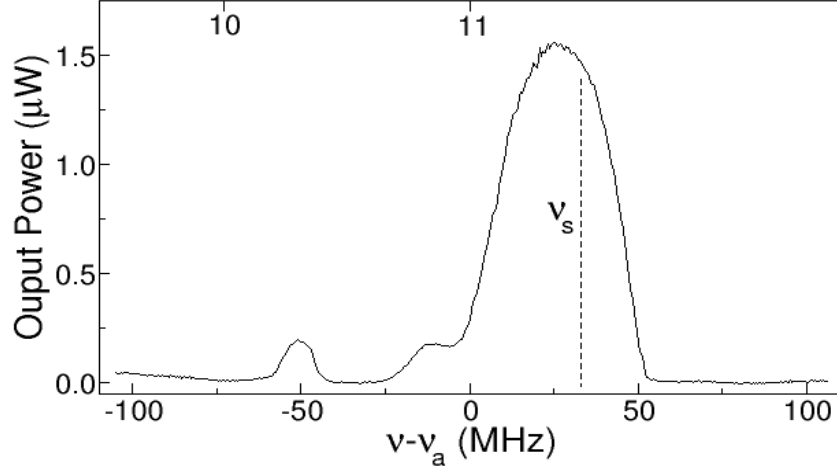


Figure S2: Resonant enhancement of the transverse optical pattern. Output power as function of the pump-beam frequency, where ν_a denotes the resonance frequency of the $5S_{1/2} (F=1) \leftrightarrow 5P_{3/2} (F'=1)$ transition and ν_s denotes the frequency at which the switching data was collected. The top-axis tick marks “10” and “11” indicate the $5S_{1/2} (F=1) \leftrightarrow 5P_{3/2} (F'=0)$ and the $5S_{1/2} (F=1) \leftrightarrow 5P_{3/2} (F'=1)$ transition frequencies, respectively.